WORKED SOLUTION

STEP II 2018, Question 12

Probability and Statistics • Optimal Stopping Expectation

STEP II 2018, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q12

The key idea

With a fixed target of h Heads, I win £h exactly when the target is reached before the game ends, and nothing otherwise. So the expected winnings are h times the probability of reaching h Heads, and the whole question is about finding that probability. To maximise over the whole number h, don't differentiate: compare consecutive terms by looking at the ratio E_{h+1}/E_h.

Solution

Throughout, let $q = 1 - p = \frac{1}{N+1}$ be the probability of Tails, and let E_h be my expected winnings with target h (where $h \geq 1$). Since I win £h if I reach h Heads and £0 otherwise,

$$E_h = h \times \mathrm{P}(\text{I reach } h \text{ Heads before the game ends}).$$

Part (i)

In version 1 a single Tail ends the game, so I reach h Heads exactly when the first h tosses are all Heads. The tosses are independent, so this has probability p^h, and

$$E_h = hp^h.$$

To find the best h, compare consecutive values. Since $E_h > 0$,

$$\frac{E_{h+1}}{E_h} = \frac{(h+1)p^{h+1}}{hp^h} = \frac{(h+1)N}{h(N+1)},$$

and

$$\frac{E_{h+1}}{E_h} \geq 1 \iff (h+1)N \geq h(N+1) \iff N \geq h,$$

with equality only when $h = N$. So

$$E_1 < E_2 < \cdots < E_N = E_{N+1} > E_{N+2} > \cdots.$$

The sequence rises strictly up to $h = N$, the next value ties with it, and after that it falls strictly. So the largest value of E_h is E_N, and choosing $h = N$ maximises my expected winnings, as required. (Choosing $h = N+1$ does equally well.) The maximum is $N(\frac{N}{N+1})^N$.

Part (ii)

Now the game ends only on two consecutive Tails. Think of the tosses up to and including the hth Head as h blocks, each ending in a Head. If I reach h Heads safely, then each block is either

$$\text{H} \quad \text{or} \quad \text{TH},$$

since two Tails in a row would have ended the game. (A Tail at the very start is fine: there's nothing before it.) Conversely, any sequence of h such blocks never contains two consecutive Tails. Each Tail in it comes straight after a Head or at the start, and straight before a Head. So I reach h Heads exactly when the tosses split into h blocks of these two kinds.

? **Ask yourself** Why can I just multiply the block probabilities together?

After each Head the game is in exactly the same position as at the start: no Tail is pending, and the future tosses are independent of the past. So each block, independently of the others, is safe with probability

$$\text{P(H)} + \text{P(TH)} = p + qp = p(1+q) = \frac{N}{N+1} \cdot \frac{N+2}{N+1} = \frac{N(N+2)}{(N+1)^2}.$$

(These two events are disjoint, because they differ at the first toss.) Hence the probability of reaching h Heads is $(\frac{N(N+2)}{(N+1)^2})^h$, and

$$E_h = \frac{hN^h(N+2)^h}{(N+1)^{2h}},$$

as required.

The case $N = 2$. Here

$$E_h = h\left(\frac{2 \cdot 4}{9}\right)^h = h\left(\frac{8}{9}\right)^h.$$

As in part (i),

$$\frac{E_{h+1}}{E_h} = \frac{8(h+1)}{9h} \geq 1 \iff 8h + 8 \geq 9h \iff h \leq 8,$$

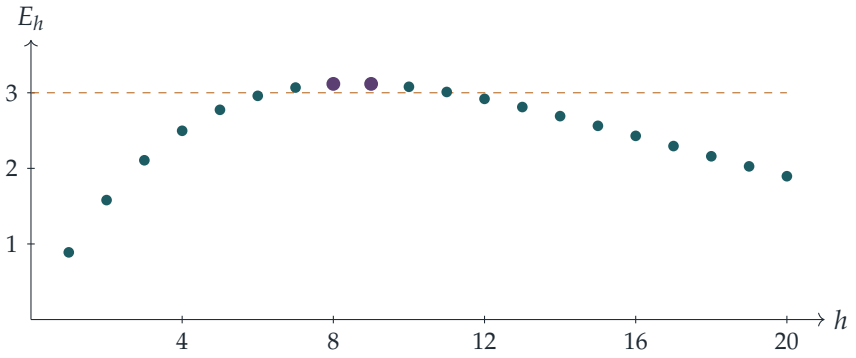
with equality only at $h = 8$. So E_h increases strictly up to $h = 8$, then $E_9 = E_8$, and then it decreases strictly. The maximum is

$$E_8 = 8\left(\frac{8}{9}\right)^8 = \frac{2^3 \cdot 2^{24}}{3^{16}} = \frac{2^{27}}{3^{16}}.$$

To estimate this, take logarithms to base 3 and use $\log_3 2 \approx 0.63$:

$$\log_3 E_8 = 27\log_3 2 - 16 \approx 27(0.63) - 16 = 17.01 - 16 = 1.01.$$

So $E_8 \approx 3^{1.01}$, which is very close to $3^1 = 3$. The maximum expected winnings are approximately £3.



The values $E_h = h(8/9)^h$ in version 2 with $N = 2$; the joint maximum at $h = 8, 9$ is just above the dashed line at 3.

Checks

- Version 2 can only be kinder than version 1, since every all-Heads start is safe in both. Indeed $\frac{N(N+2)}{(N+1)^2} = 1 - \frac{1}{(N+1)^2} > \frac{N}{N+1}$.
- With $N = 1$, version 1 gives $E_1 = E_2 = \frac{1}{2}$: one Head with probability $\frac{1}{2}$, or two with probability $\frac{1}{4}$.
- Exactly, $E_8 = 2^{27}/3^{16} = 134\,217\,728/43\,046\,721 \approx 3.118$, so the estimate £3 is good. A computer simulation of 2×10^5 games with $N = 2$ and $h = 8$ gave average winnings of about £3.11.

Common slips

- In version 2, forgetting that a single Tail at the start is allowed, or wrongly thinking a Tail may follow the hth Head. I stop as soon as the hth Head appears.
- Differentiating hp^h and setting the result to zero. This gives $h = -1/\ln p$, which usually isn't an integer, and turning it into the best integer needs care. The ratio test is cleaner and exact.
- Missing the tie: $h = N$ and $h = N + 1$ give the same expected winnings, so the "show that" asks only that $h = N$ achieves the maximum.

Beyond STEP

This is a simple *optimal stopping* problem. Here I fix the target in advance, but you can check that nothing is lost by doing so: whenever I hold H Heads with no Tail pending, the gain from one more toss is positive exactly when H is small enough. Problems like this, where one decides when to stop by comparing "stop now" with "one more step", lead to martingales and dynamic programming. The block argument in part (ii) is a first taste of renewal ideas for Markov chains. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

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